Mapping to a reference genome

Yesterday: Quality check and trimming raw reads

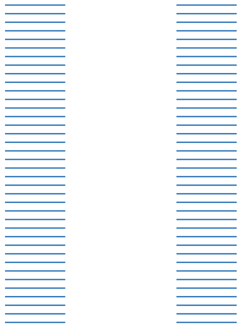
paired-end Illumina sequencing

Up to 20 billion read pairs

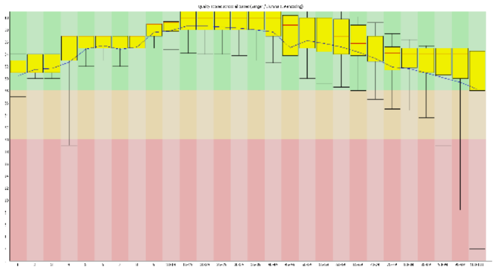
150 bp 150 bp

Forward reads

Reverse reads

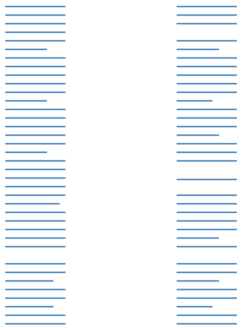


Fastqc quality plots

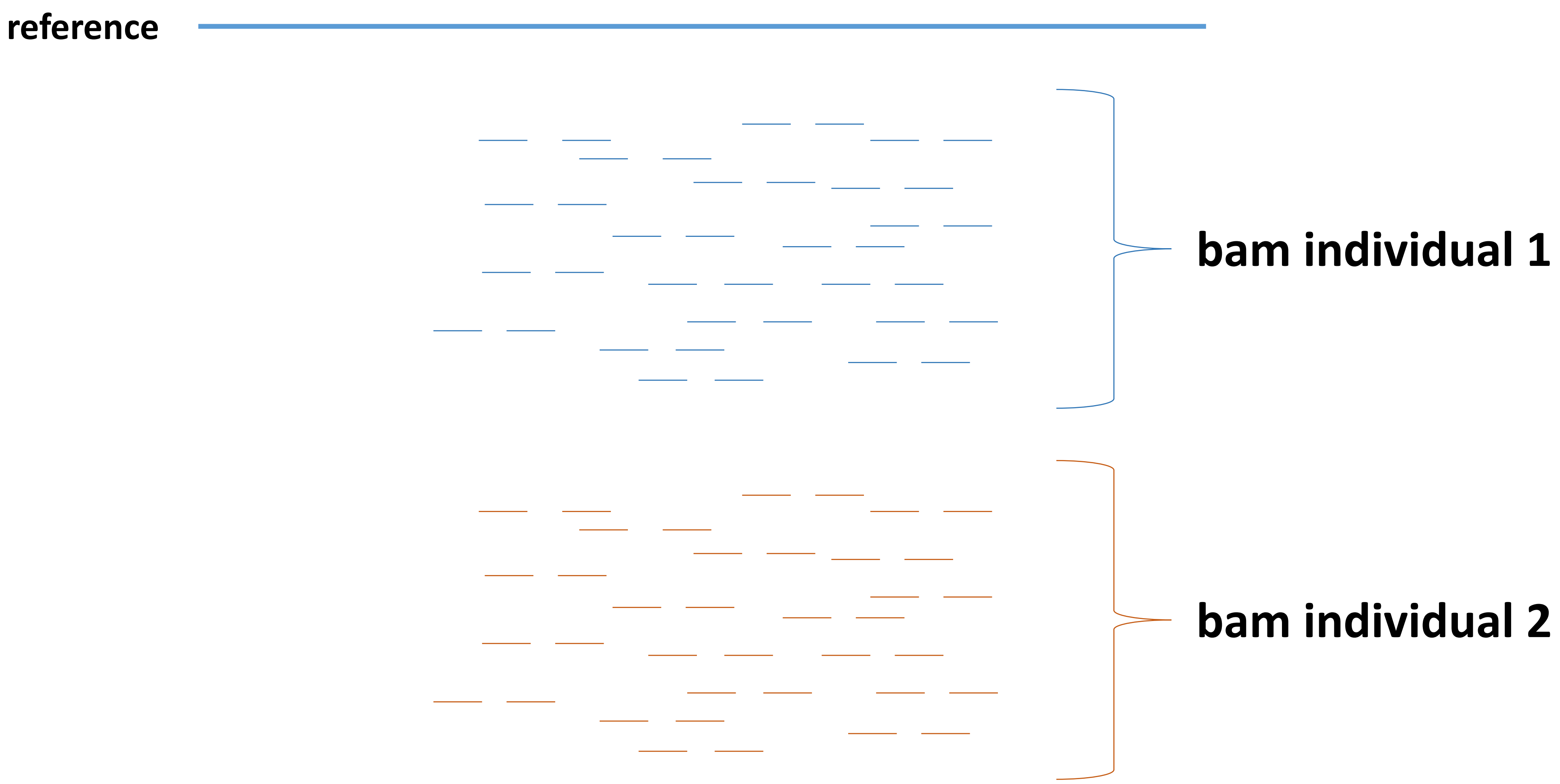


fastp

Trimmed and filtered reads of good quality



Today: Alignment to the reference genome



Aligning/Mapping short read data to a reference genome

bwa-mem2 (Burroughs-Wheeler Alignment with maximum exact matches)

Version 2 is 50% faster than bwa-mem

Very new tool worth considering: minibwa (https://arxiv.org/abs/2606.15357) which is much faster, similar quality and even better at aligning reads with indels

Regions that are repeated in the genome make it very difficult to map reads

Reference genome

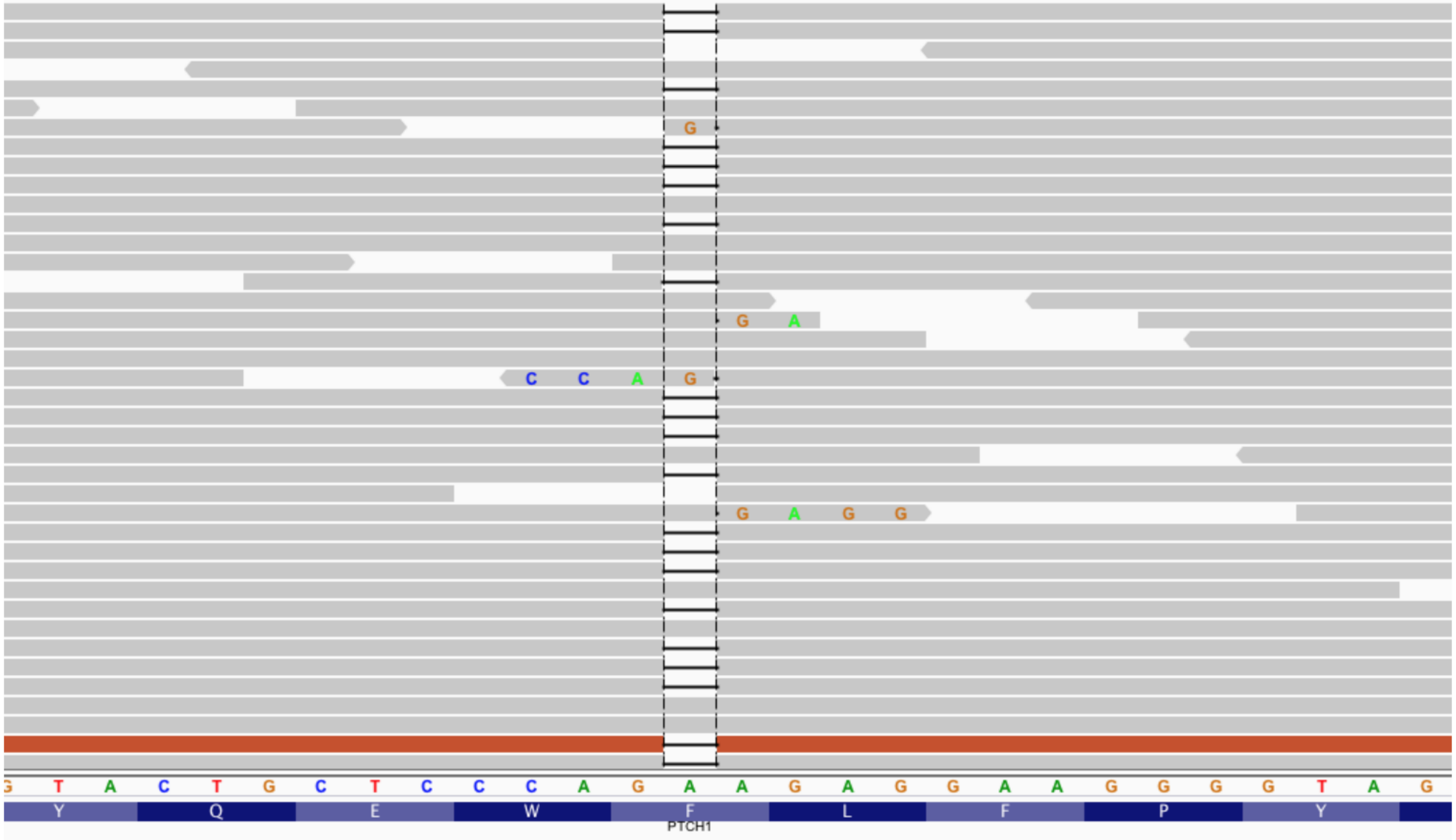
Read to be aligned to the reference

Regions that are repeated in the genome make it very difficult to map reads

Reference genome

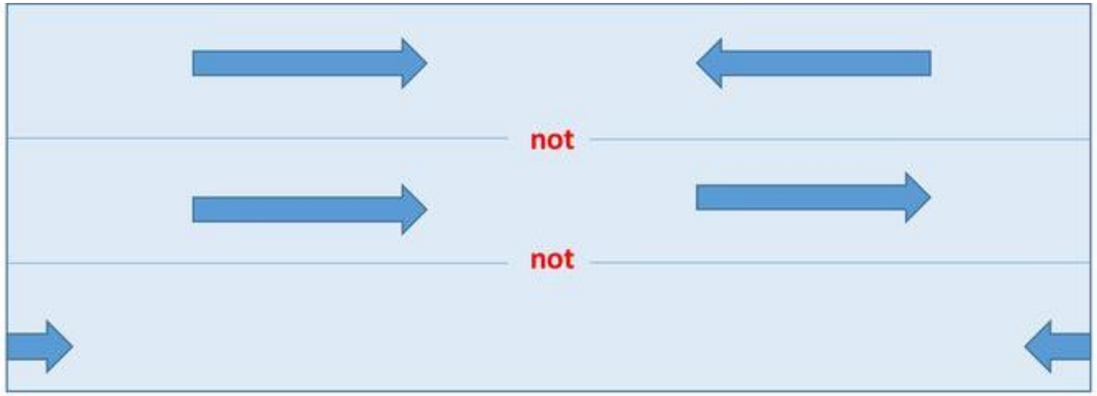
Read to be aligned to the reference

Indels: deletions and insertions



Aligning reads is complicated by:

- Imperfect match to the reference due to mutations or sequencing errors, or errors in the reference genome
- Multiple positions where the read could match (repeated regions)
- Low quality of the read
- With paired-end reads: only one read maps or the other read maps on a different chromosome



Alignment tools such as bwa-mem2 are able to handle all of these complications and give information on the mapping quality.

General principle

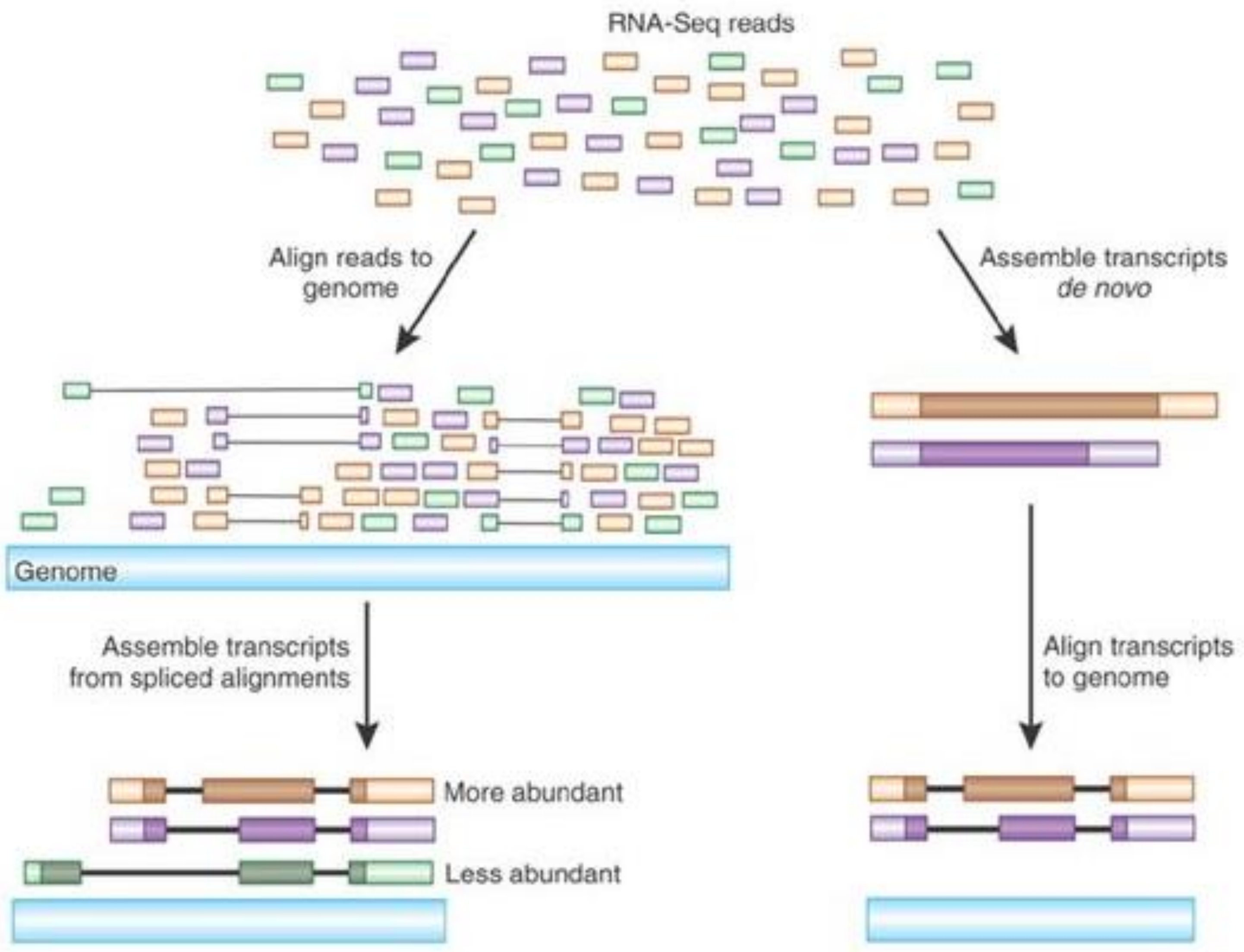
- The reference genome is indexed to make mapping faster
  - Reference: ATCATTCATCTG (would of course be much longer)
  - Index: for each k-mer figure out where they are in the genome
    e.g. for k=3, k-mers are combinations of 3 nucleotides

    Positions of 3-mers in our toy example:
    ATC: 1,8 (**ATC**ATTC**ATC**TG)
    TCA: 2 (A**TCA**TTCATCTG)
    CAT: 3,7 (AT**CAT**T**CAT**CTG)
    ATT: 4 (ATC**ATT**CATCTG)
    etc.

- The reads are mapped using the index:
  - Read example: CATTC
  - First k-mer (CAT) suggests the read could start at position 3 or 7
  - Second k-mer shows that it matches (ATT) at position 4

Aligning other type of data: RNA data

Tools:
e.g.
STAR
HISAT2



Haas et al. 2010

Checking if a genome exists: https://goat.genomehubs.org

GoaT - latest   search   help   projects

Genomes on a Tree (GoaT)

> **Genomes on a Tree (GoaT): A versatile, scalable search engine for genomic and sequencing project metadata across the eukaryotic tree of life**. Challis *et al.* 2023. Wellcome Open Res 2023, **8**:24 doi:10.12688/wellcomeopenres.18658.1

GoaT has been built using GenomeHubs to help coordinate efforts across the Earth Biogenome Project (EBP) Network at all stages from planning through sequencing and assembly to publication. read more...

Search GoaT

Type to search GoaT taxon index (e.g. Canidae)   TAXON

include descendants Off | include estimates On | empty columns Off | result columns | query builder | clear all

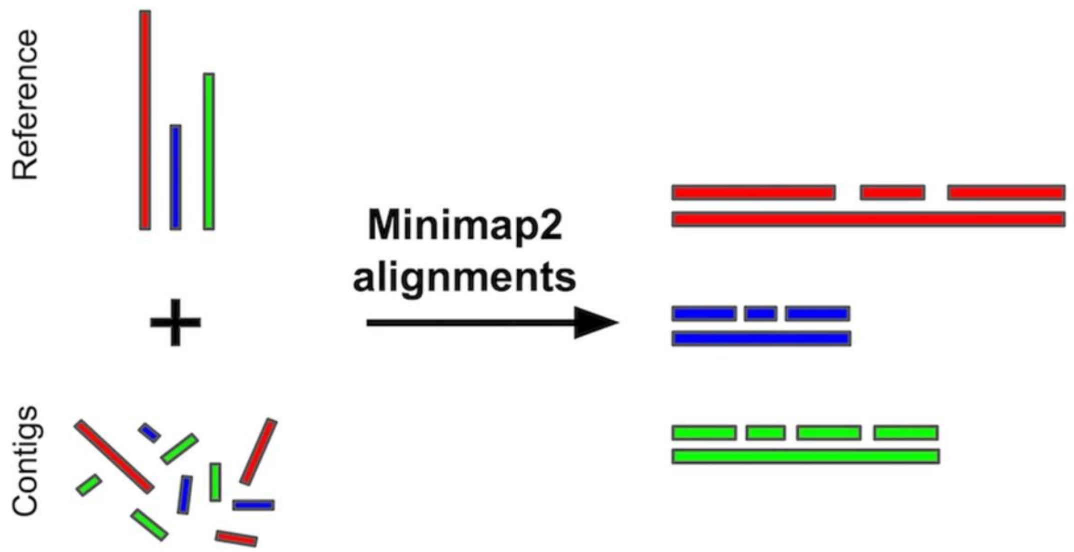
Try the new search box   show examples or browse tree

Improving a draft genome assembly with a chromosome-level reference genome of a close relative

Reference-guided scaffolding with RagTag (successor of RAGOO)

https://github.com/malonge/RagTag

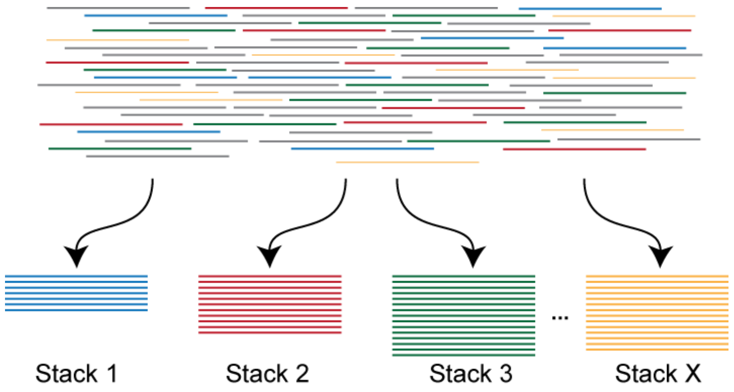
Uses minimap2 for genome alignment



RAD data without a reference genome

Make your own reference with:

- **STACKS**: for datasets of
  https://catchenlab.life.illinois.edu/stacks



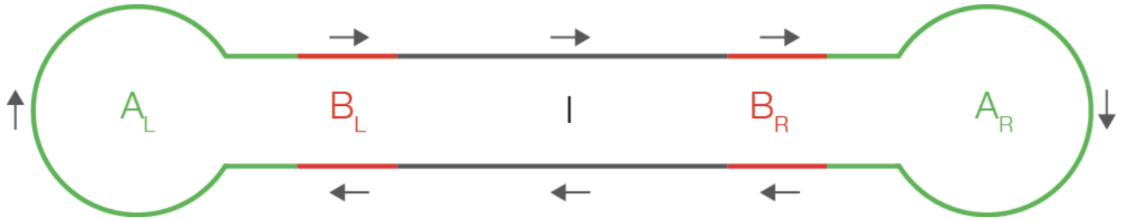
Stacks

- **iPyRAD** (for RAD, ddRAD, GBS, 2bRAD): for datasets with different species
  https://ipyrad.readthedocs.io/en/master/index.html

Long read mapping

Karin Näsvall

Mapping long reads PacBio



A schematic drawing of a SMRTbell

A =Adaptors, B=Barcode, I=insert (DNA of interest)

Automatic HiFi reads generation (Export Reads)

`hifi_reads.fastq.gz` - FASTQ file containing HiFi Reads
`hifi_reads.fasta.gz` - FASTA file containing HiFi Reads
`hifi_reads.bam` - BAM file containing HiFi Reads

HiFi reads QV > 20

https://pacbiofileformats.readthedocs.io/en/13.0/

PacBio

cat m84093_240426_124306_s1.ccs.bc2033.stats

-

A = 7108650862 (34.9%), C = 3058767823 (15.0%), G = 3061827434 (15.0%), T = 7127816130 (35.0%), CpG = 1003156910 (4.9%)

sum = 20357062249, n = 2218131, mean = 9177.57438537219, largest = 30301, smallest = 135

sum = total number of bases

n = number of sequences

mean, largest, smallest = length of reads

PacBio

https://pacbiofileformats.readthedocs.io/en/13.0/BAM.html#hifi-reads

m84093_240426_124306_s1/261297081/ccs 4 * 0 255 * * 0 0
GTTTGGTTATATTGTTCATAGACACCTACGGTACACGAATATAATAAACAAATAATATTAAATTGTTATACATATTTAAATATAAATTAATATTGTAGCAACTATTAACTAACTAGTTCAC
TTCCTTCAACAGTTTAAATCAAGGATTCCTTAACTATTTTTTACCAAAACGTACTGATACCAAAGATTGTTTTCTAATGATTTTGATGATATGGATGTGATTTGACTACATTAATGCTA
CTGCTAGAACTTTACGATATCCCTCACATCACCTCGTCAAAATGCTTTGTTTAAATTGTTAAAAAAAATTCTTTGTATTTAAAGTGAGTAATCCTTAAGTAAACAGTGTTTCAAAAGT
TTAAAATTTATTTTATACGTGAAATGTCGTATTATCGATGGCATTATTAGTTTCCAGCATATTTATAAAAGCCTTTGTGACGCAATCACCGAGCGGTGAAAATATGATTTGAGACGCT
TCCTGAAATTGAAAAAAGAAACCCTCTATTGTTTTTAGATGATGTTTATTTTCTGCATTACTTTCTTAAAATTAATATAATTTATACTATACAAATGTGACATTTCAATTGCGATACGA
GTTTAAGAGGAGAACATCGATTACTGAAGTTATACAAACTTCTCCACAATAGATAACATTAACTTTATATTTTAAACTTTAAGATAGTTCTTAAAAGTGTGAAAAGATGTAAAAAAT
GGTACATTTGAAGAAACCTTTATTATGAAAGAGTTTAAAAGGTATCGGTATTTTCAAAAAGATTTTACATCTTTCTAATGCGAAATGTTAAAAATCTTAAAAATATTTTTTAGCTTAT
TTAACGAATAAGAAGAGAAAATATTTATTAAAATTCATATTTGATATATTTATATTATTGATTTTATTTATTCCTTTTTAAATAGTTTTAATTTAGTTCGATTTCGAAATCTCTTTTATTGT
TAAATGTTTCATCTGGAGACATTTACAGTCCAATAGTTTCCAAAAGAAATTTTAAAACGCTGTTTCGAAATTATCTTTCGTGATTATGAACTAGTTAAAAATTCCACTCAGTTAATG
GTTTAAAATTTTATGTATTGCAAGAAAAGATAATAAAAGCTTGAAATAAAATCATTTTATCAGTATTTTGCATACCTATAAATATTTGAAGGTA

IIIIIIIIIIIIDIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIDIIIIIIIIIII7IIIIIIIIIIIIIIIIIIIIIIIIIIIIIII<IIIIIDIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIDIIIII
IIIIIIIIIIIIIIIII+IIIIIIIIIIIIIIIIIIIIIIIIIIIIIII+III7IIIIIIIIIIIII+IIIIIIIIIIIIDIIIIIIIIIIIIIIIIIIIIIIIIIIIDIIIIIIIIIIIIII2IIIIIIIIIIIIIDI2IIIIIIIIIIIIIIIIIIII7IIIIIDIIIIIIIIIIIIII

ac:B:i,8,0,10,0 bl:Z:CGACTCGTATGAGTAT bt:Z:ATACTCATACGAGTCG bx:B:i,16,16 ec:f:7.5012 ma:i:0 np:i:7 ql:Z:IIIIIIIIIIIIIIII qt:Z:IIIIIIIIIIIIIIII rq:f:0.999815
sn:B:f,17.2518,25.3332,6.55462,11.2707 we:i:8638323 ws:i:40959 zm:i:261297081 qs:i:26 qe:i:17132 bc:B:S,0,0 bq:i:100 cx:i:12
ls:B:C,134,162,98,99,164,48,45,45,48,162,98,113,100,164,108,101,97,100,130,162,113,108,186,73,73,73,73,73,73,73,73,73,73,73,73,73,73,73,73,73,73,73,
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,162,115,113,186,65,71,84,65,67,84,67,84,71,67,71,84,84,71,65,84,65,67,67,65,67,84,71,67,84,84 RG:Z:f56a67e5/0--0

PacBio mapping – Minimap2

Minimap2 https://github.com/lh3/minimap2

Uses **minimizers** – short sequences of length –k [default 15 bases]

1. Extracts minimizers from the reference (target) and index them
2. Match each minimizer in the query sequence against the reference set of minimizers
3. Sorts the minimizers after genomic position
4. Make a chain of overlapping minimizers
5. Repeat for all query sequences

minimap2 –ax **map-hifi** ref.fa query.bam > output.sam # Pacbio HiFi

minimap2 -ax **map-ont** ref.fa ont.fq.gz > output.sam # Oxford Nanopore

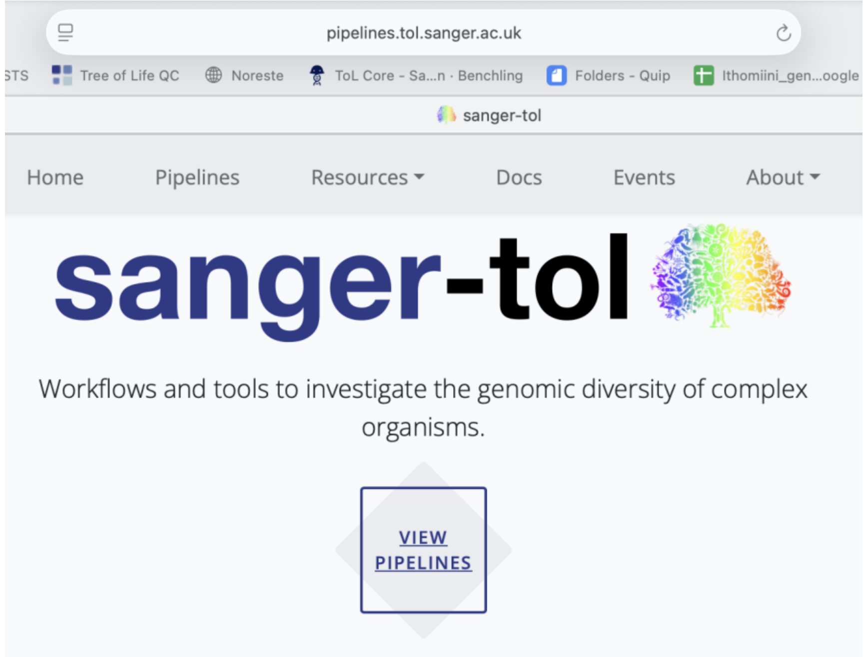
Readmapping pipeline

nextflow

Preprocessing - filtering and adaptor removal

Alignment - minimap2

Alignment postprocessing - mapping statistics



https://pipelines.tol.sanger.ac.uk

Generating and finding reference genomes

Generating reference genomes with long reads and Hi-C data

Contigs from assembled long reads

Contig 1

ATAGTGTGGATGCTGAGTTCGT

Contig 2

TTCAGGGGCATAAAAGTAACGTA

Contig 3

GACACCCGCATATTAGTAAC

Contig 4

ATCGTA...

Hi-C data

ATAGTG — CTGAGT

TTCGT — CAGGGG

AGTTC — CATAAA

GGCAT — ACGTA

GACAC — AGTAAC

GCATAT — ATCGTA

Scaffold 2

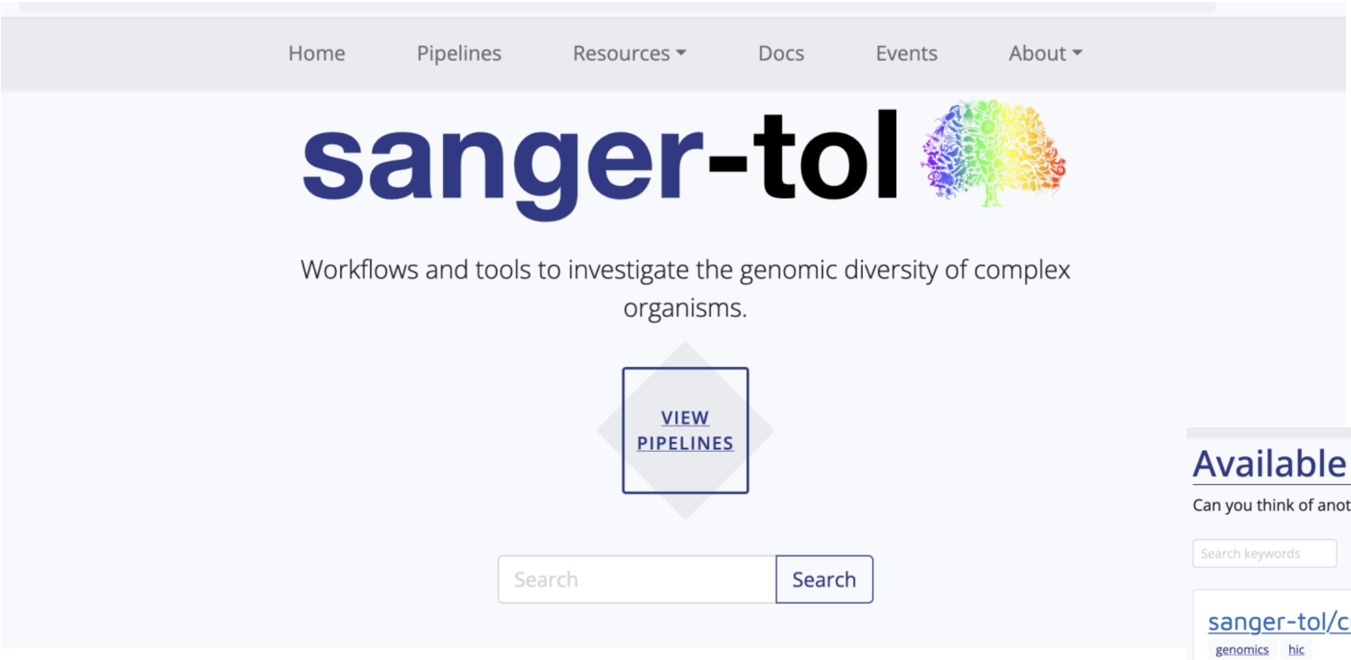
ATAGTGTGGATGCTGAGTTCGT **NNN** TTCAGGGGCATAAAAGTAACGTA

Scaffold 2

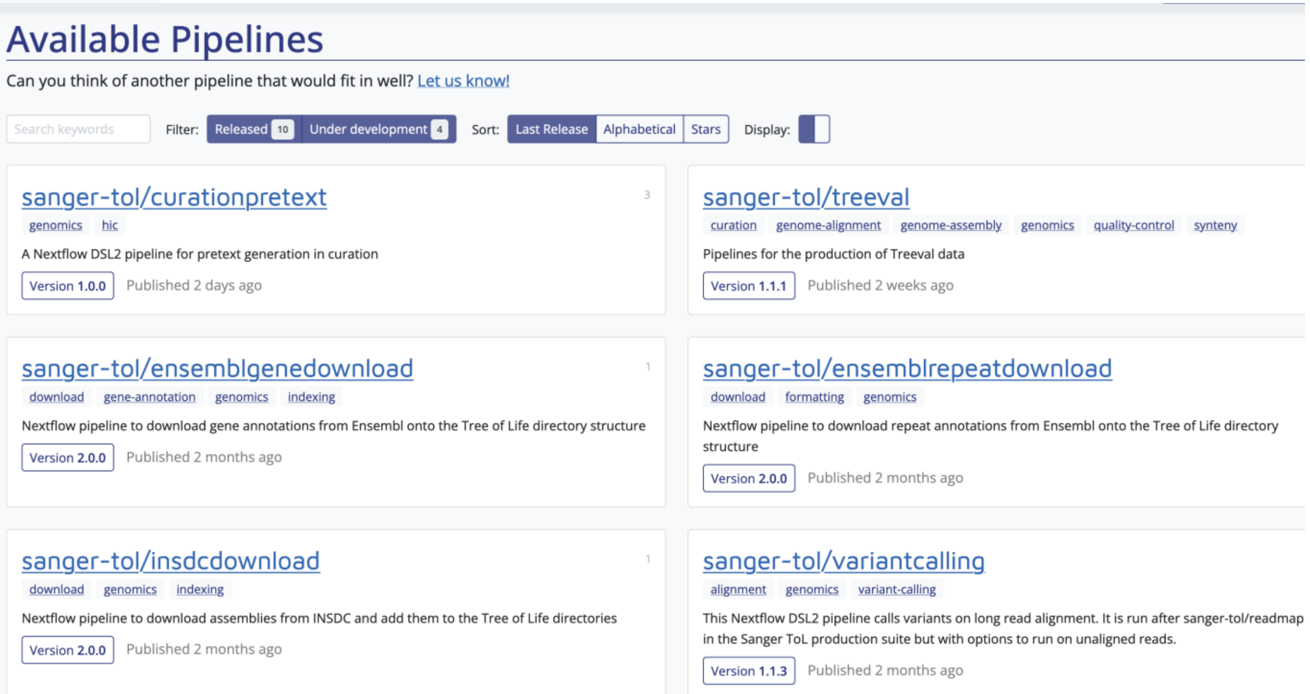
GACACCCGCATATTAGTAAC **NNN** ATCGTA...

Pipelines to generate a genome

https://pipelines.tol.sanger.ac.uk/



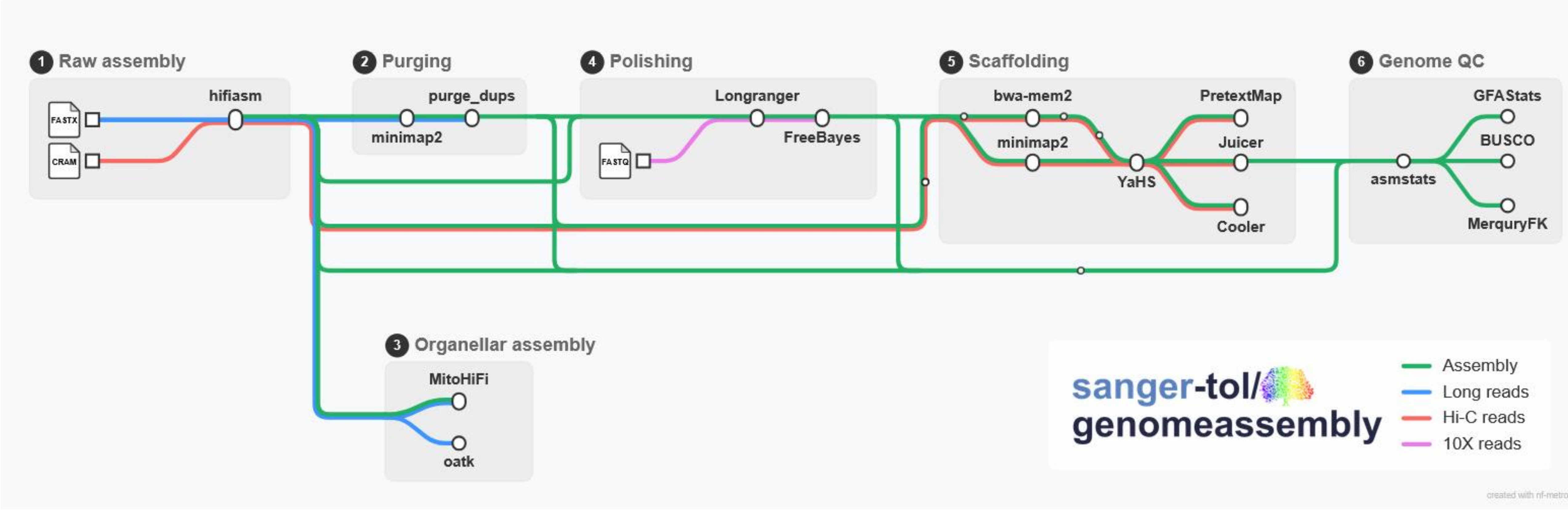
Sanger-Tol recipe is working across the Tree of life
>4000 species already with reference genomes



Chromosome level genomes

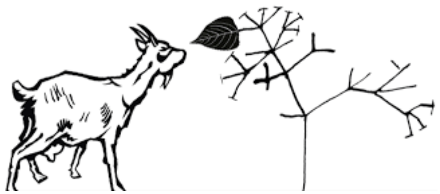
- 25x Pacbio HiFi
- 100x Hi-C (Arima/Qiagen)

Tree of Life pipeline for genome assembly



Checking if a species or genus has a genome

https://goat.genomehubs.org



GoaT - latest | search | help | projects

Genomes on a Tree (GoaT)

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> Challis *et al.* 2023. Wellcome Open Res 2023, **8** doi.12688/wellcomeopenres.18658.1

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Search GoaT

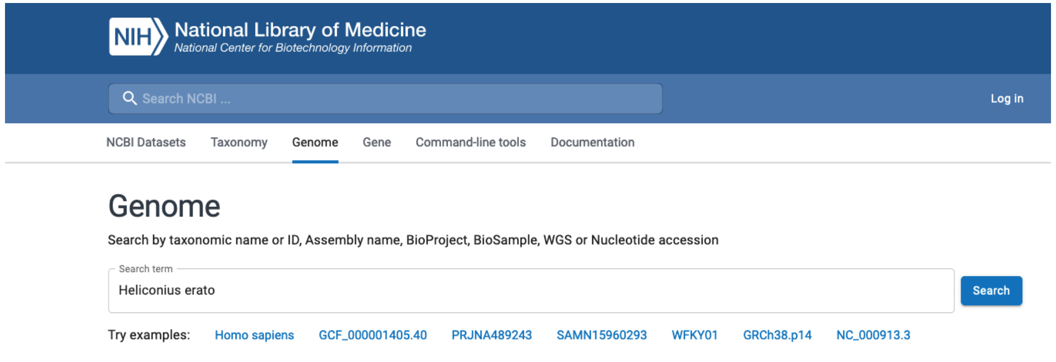
Type to search GoaT taxon index (e.g. Canidae) TAXON

include descendants Off | include estimates On | empty columns Off | result columns | query builder | clear all

Where can we find genomes?

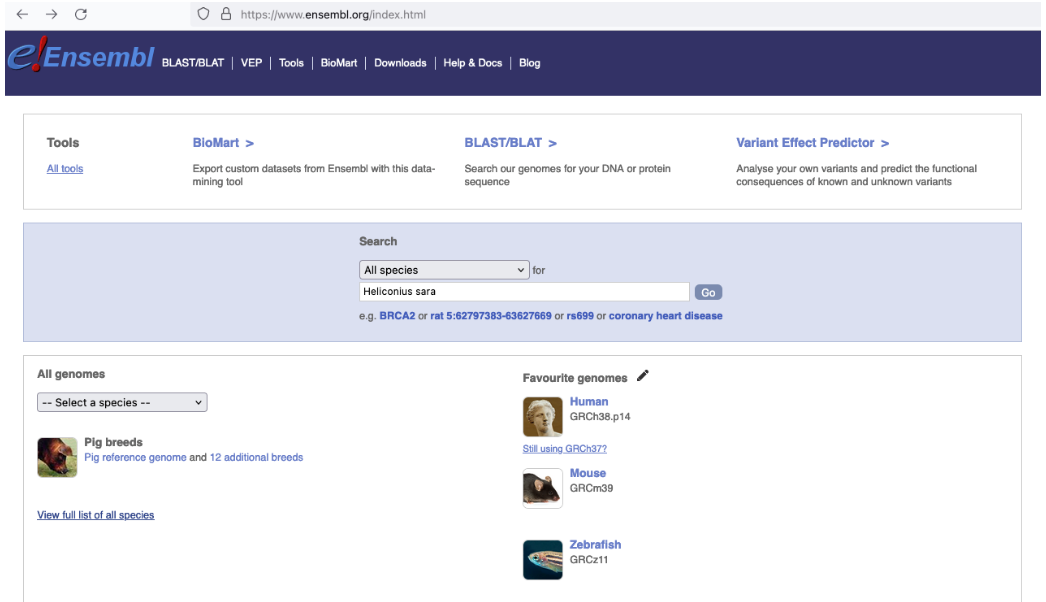
- NCBI (National Center for Biotechnology Information):

https://www.ncbi.nlm.nih.gov/datasets/genome/



- ensembl

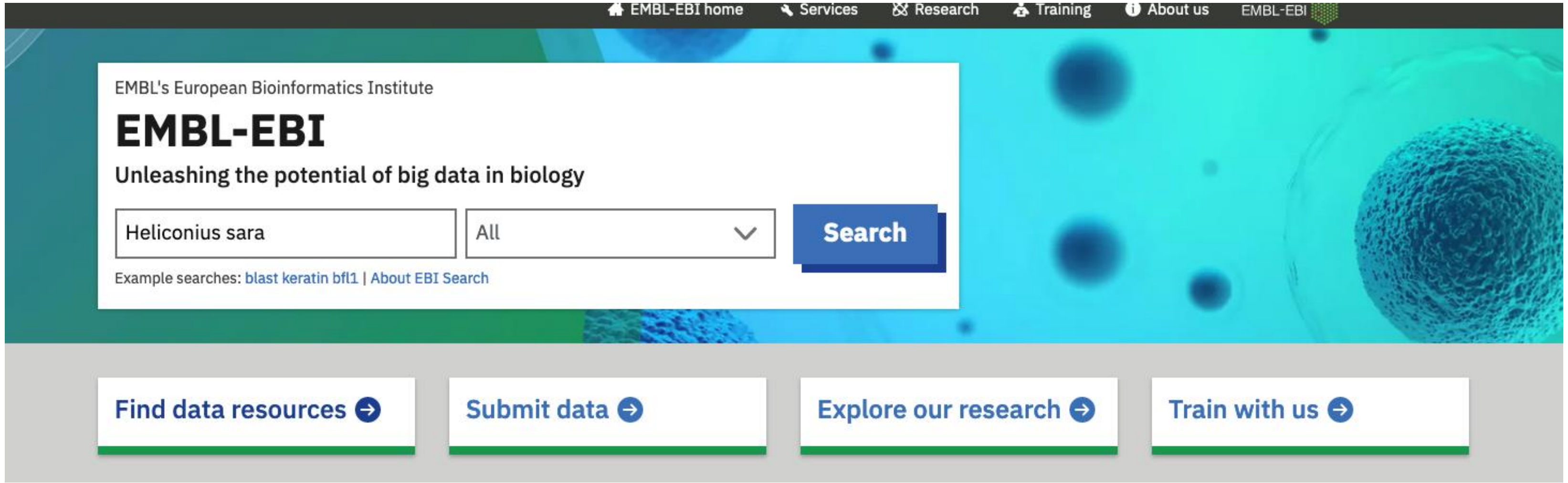
https://www.ensembl.org/index.html



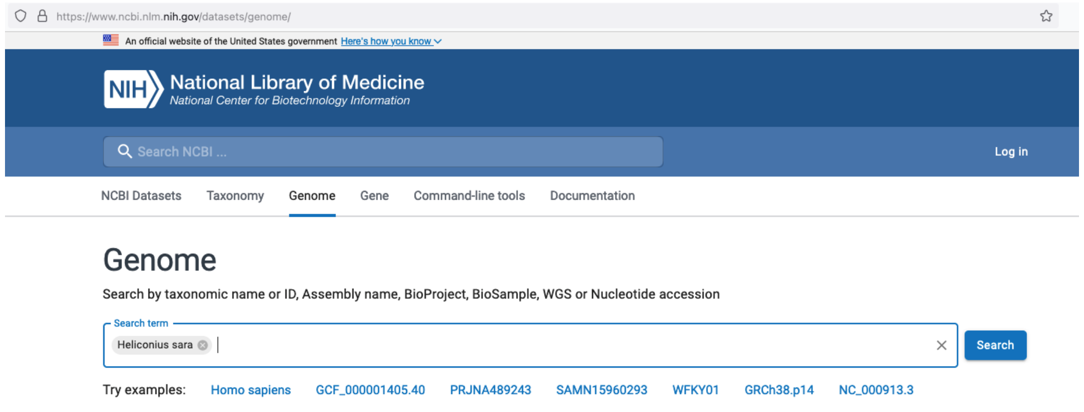
Where can we find genomes?

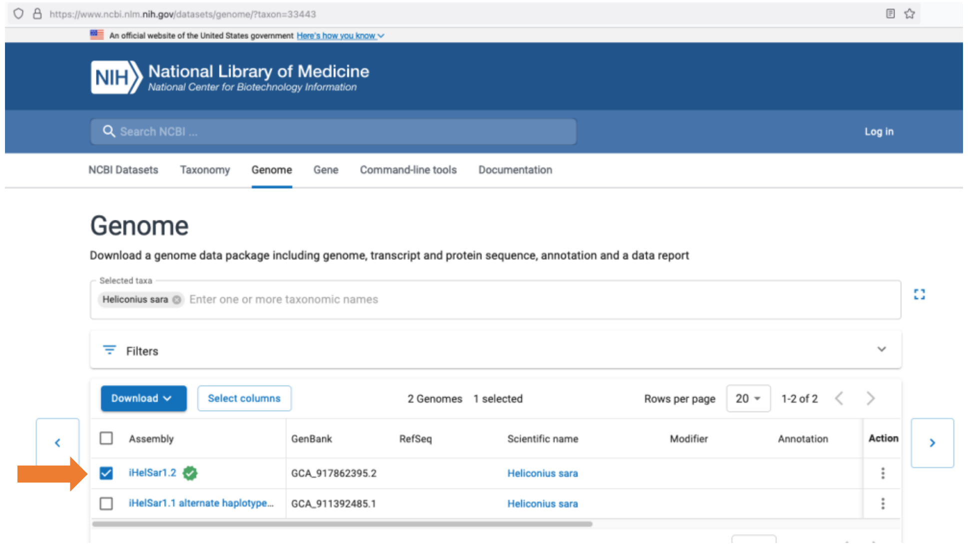
- EMBL-EBI (European Molecular Biology Laboratory - European Bioinformatics Institute):

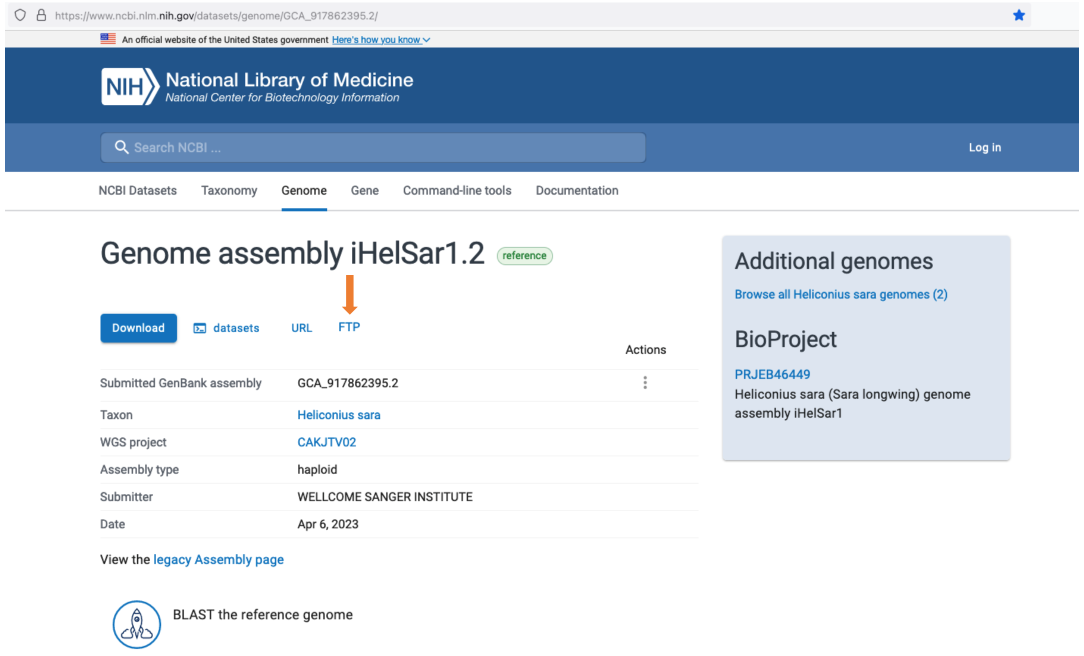
https://www.ebi.ac.uk/

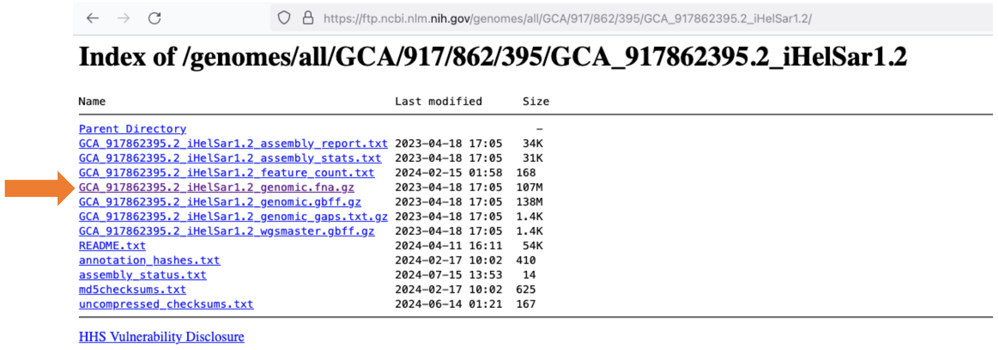


Example










```
wget
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/917/862/395/GCA_917862395.2_iHelSar1.2/GCA_917862395.2_iHelSar1.2_genomic.fna.gz
```

Reference genomes are stored in a fasta format

It is a text-based format for representing nucleotide sequences (DNA or RNA)

```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)
ACCCTCTTTTCTTATCATTGACATTTAAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCGGGCTCCGGCCCCGGCCCGGCTCGGGGCCCGCGGGGCCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCGCCCCAAGTGGCCCCGGGGCTTGATTTTTGCTTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGACTTGTCTT
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGGTCGGGCGAGGTGCAGCAGAAGTTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

Description Line

Sequence Lines

Description Line: Begins with a > character followed by an identifier and optional description.

Sequence Lines: The actual sequence data, typically represented over multiple lines for readability.

Assembly statistics

	GenBank
Genome size	562.2 Mb
Total ungapped length	562.2 Mb
Number of contigs	203,084
Contig N50	9.9 kb
Contig L50	14,088
GC percent	33.5
Genome coverage	1.9x
Assembly level	Contig

It is the median of the contig lengths.

It is the number of contigs (or scaffolds) that together contain at least 50% of the total assembly length.

Significance

N50: A higher N50 value indicates that the assembly contains longer contigs, suggesting better assembly quality and continuity.

L50: A lower L50 value indicates that fewer contigs are needed to cover 50% of the genome, suggesting higher assembly contiguity.